LAB # 10

Yusuf Bozdağ

Student ID #: 33840

Lab Section : C

1. Introduction

The purpose of this experiment is to study the operation and characteristics of MOSFET amplifiers using a BS170 NMOS transistor. In this lab, a common-source amplifier configuration was designed, analyzed, simulated, and implemented on a breadboard. The experiment focused on determining important transistor parameters such as the threshold voltage and conduction parameter, as well as examining the DC operating point, voltage gain, and frequency response of the amplifier.MOSFETs are widely used in analog and digital electronics because of their high input impedance, low power consumption, and amplification capabilities. In a common-source amplifier, the input signal is applied to the gate terminal while the output is taken from the drain terminal. This configuration provides voltage amplification together with a phase inversion between the input and output signals. During the experiment, theoretical calculations were first performed to estimate the Q-point and small-signal gain of the amplifier. LTspice simulations were then used to verify the expected circuit behavior before implementing the circuit experimentally. Finally, measured results obtained from the oscilloscope and multimeter were compared with theoretical and simulation values in order to evaluate the performance of the amplifier and understand practical non-ideal effects in MOSFET circuits.

2. Pre-Lab Work

The purpose of this pre-lab is to analyze the common-source NMOS amplifier before simulating it in LTspice. In this circuit, the BS170 NMOS transistor is biased with a voltage divider at the gate, a drain resistor, and a source resistor. The main goals are to find the DC operating point, calculate the terminal voltages, draw the DC load line, estimate the midband voltage gain, and determine the lower and upper cutoff frequencies.

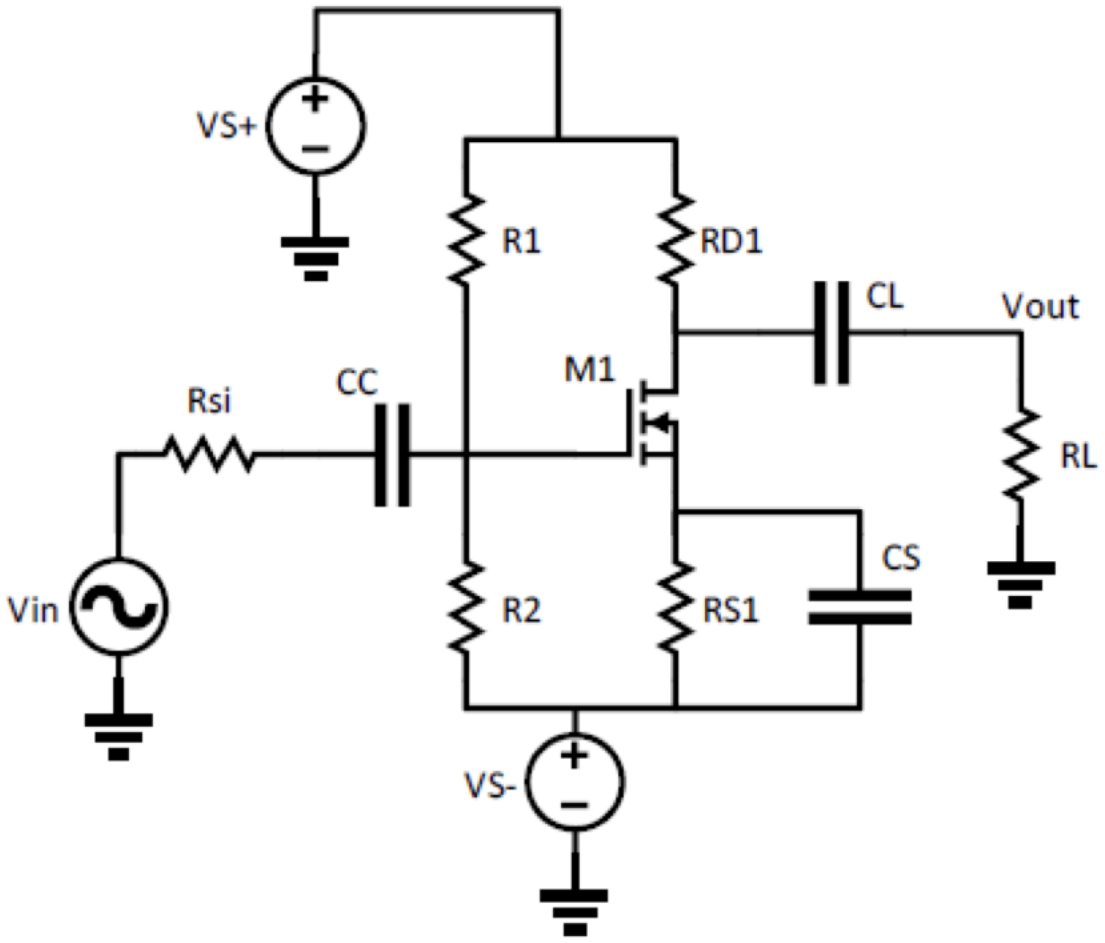


Figure 1

The circuit uses +5 V and −5 V supplies. The given resistor values are R1 = 200 kΩ, R2 = 100 kΩ, RD = 1 kΩ, RS = 240 Ω, and RL = 1 kΩ. The coupling capacitors and the source bypass capacitor are all 100 µF. The transistor model is the BS170 VDMOS model given in the pre-lab sheet. For the hand calculations, the most important parameters are the threshold voltage, Vt = 2.08 V, and the transconductance parameter, KP = 0.1233. The calculations below follow the same approach as the handwritten solution, where the gate bias is first found from the voltage divider and then the transistor operating point is calculated.

For the DC analysis, all capacitors are treated as open circuits. This means the input source is blocked by the input coupling capacitor, the load is blocked by the output coupling capacitor, and the source bypass capacitor does not affect the DC bias. Therefore, the DC operating point is determined only by R1, R2, RD, RS, and the transistor.

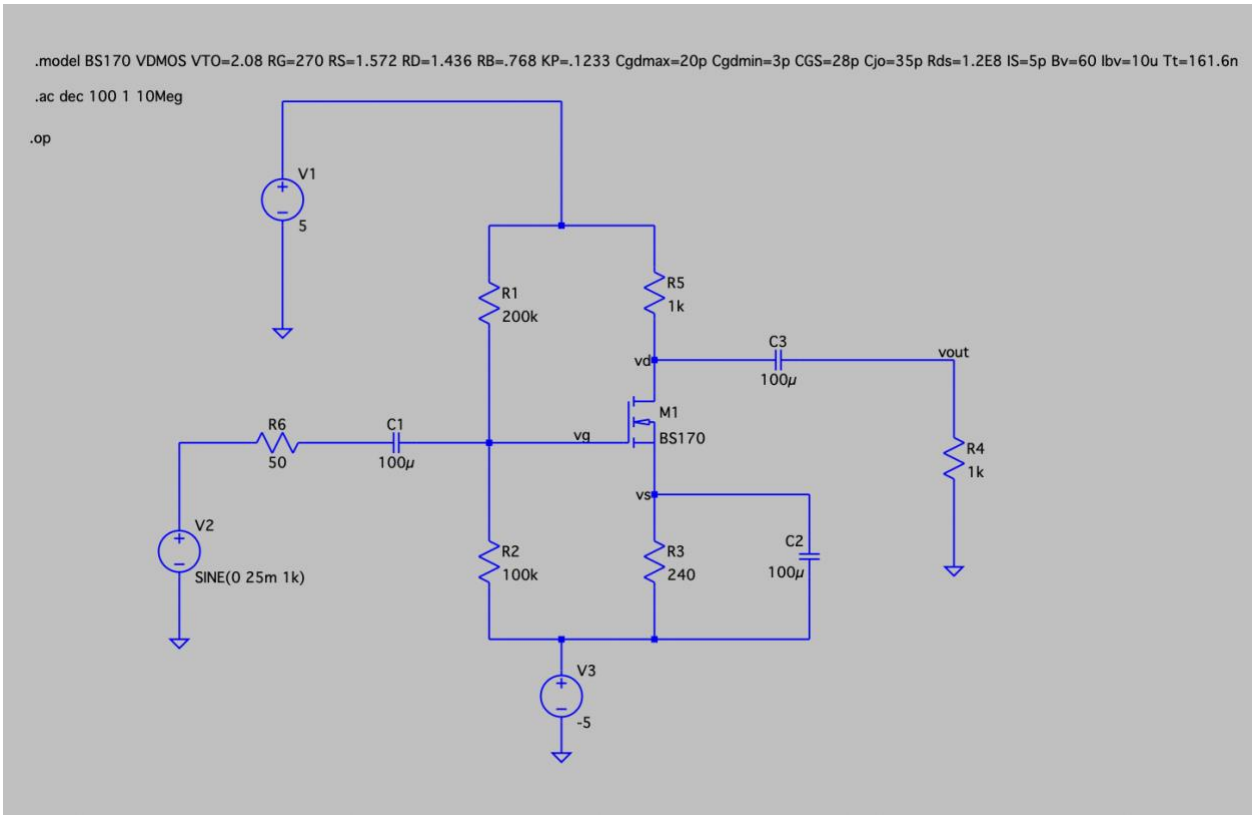


Figure 2. Ltspice Schematic

First, the gate voltage is calculated using the voltage divider between +5 V and −5 V. The total voltage across R1 and R2 is 10 V. Since R1 + R2 = 300 kΩ, the current through the divider is 33.33 µA. The voltage drop across R1 is 6.67 V, so the gate voltage is:

VG = 5 V − 6.67 V = −1.67 V

Therefore, the gate is biased at approximately −1.67 V with respect to ground.

Next, the source and drain voltages are written in terms of the drain current. Since the source resistor is connected to −5 V, the source voltage is: VS = IDRS − 5

Using RS = 240 Ω: VS = 240ID − 5

The drain resistor is connected to +5 V, so the drain voltage is: VD = 5 − IDRD

Using RD = 1 kΩ: VD = 5 − 1000ID

The gate-source voltage is found from VG − VS. Since VG = −1.67 V, this gives: VGS = −1.67 − (240ID − 5) VGS = 3.33 − 240ID

Using the BS170 threshold voltage of 2.08 V, the MOSFET current equation is solved. This gives two mathematical roots, but only the physically meaningful solution is used because the other one does not satisfy the expected NMOS operating condition. The valid drain current is: IDQ ≈ 4.43 mA

Using this current, the source voltage becomes: VS = (4.43 mA)(240 Ω) − 5 = −3.94 V

The drain voltage becomes: VD = 5 − (4.43 mA)(1 kΩ) = 0.568 V

Then the gate-source voltage is: VGS = VG − VS = −1.67 − (−3.94) = 2.27 V

The drain-source voltage is: VDS = VD − VS = 0.568 − (−3.94) = 4.5 V

So, the DC operating point of the amplifier is approximately: IDQ = 4.43 mA and VDSQ = 4.5 V

The terminal voltages with respect to ground are: VG = −1.67 V, VS = −3.94 V, and VD = 0.568 V

After finding the Q-point, the saturation condition should be checked. For an NMOS transistor, saturation requires VDS to be greater than VGS − Vt. Here, VGS − Vt is 2.27 − 2.08 = 0.19 V. Since VDS is 4.5 V, the condition is clearly satisfied. Therefore, the transistor is operating in the saturation region. This is important because the common-source amplifier is designed to work properly in saturation.

The DC load line is found by writing the voltage relation between the two supplies through RD, the transistor, and RS. Since the total supply difference is 10 V, the load line can be written as: VDS = 10 − ID(RD + RS)

Substituting RD = 1 kΩ and RS = 240 Ω gives:VDS = 10 − 1240ID

This equation gives the DC load line. When ID = 0, VDS = 10 V. When VDS = 0, the current is 10 / 1240 = 8.06 mA. Therefore, the load line should be drawn between the points (10 V, 0 mA) and (0 V, 8.06 mA). The Q-point should be marked on this line at approximately VDSQ = 4.5 V and IDQ = 4.43 mA.

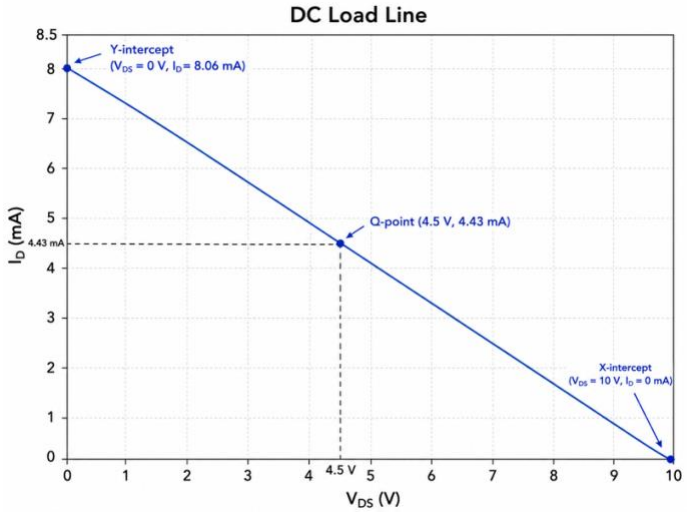


Figure 3. DC Load Line

For the small-signal analysis, the circuit is considered in the midband frequency range. In this region, the coupling capacitors and the source bypass capacitor behave almost like short circuits. Because the source resistor is bypassed by CS for AC signals, the source terminal becomes approximately AC ground. This increases the voltage gain of the amplifier.

The voltage gain of a common-source amplifier is approximately equal to −gm times the effective drain resistance. The effective drain resistance is RD in parallel with RL. Since both RD and RL are 1 kΩ, their parallel combination is 500 Ω.

Using the small-signal calculation from the pre-lab notes, the transconductance is approximately 31.3 mA/V. Therefore, the voltage gain is: Av ≈ −(31.3 mA/V)(500 Ω) Av ≈ −15.6 V/V

The negative sign means that this amplifier inverts the input signal. In other words, when the input signal increases, the output signal decreases, and the output is approximately 180 degrees out of phase with the input. This gain value is also consistent with the LTspice AC simulation, where the midband gain is around 15 V/V.

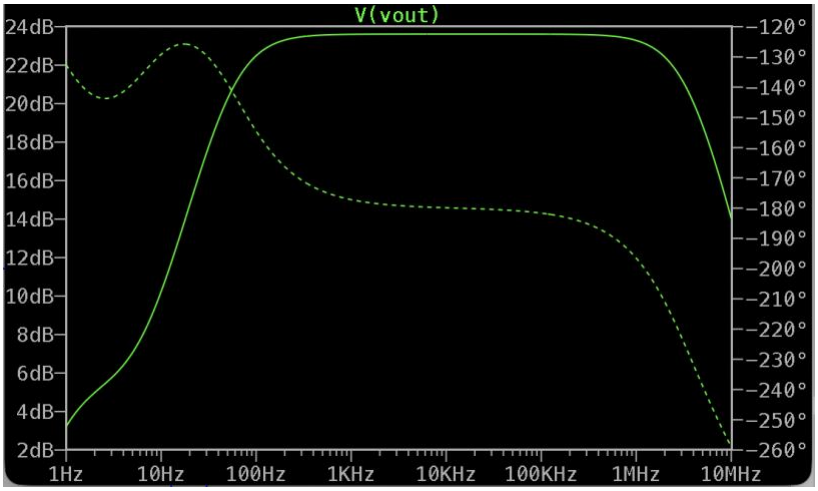


Figure 4. LTspice AC analysis result shown in dB scale.

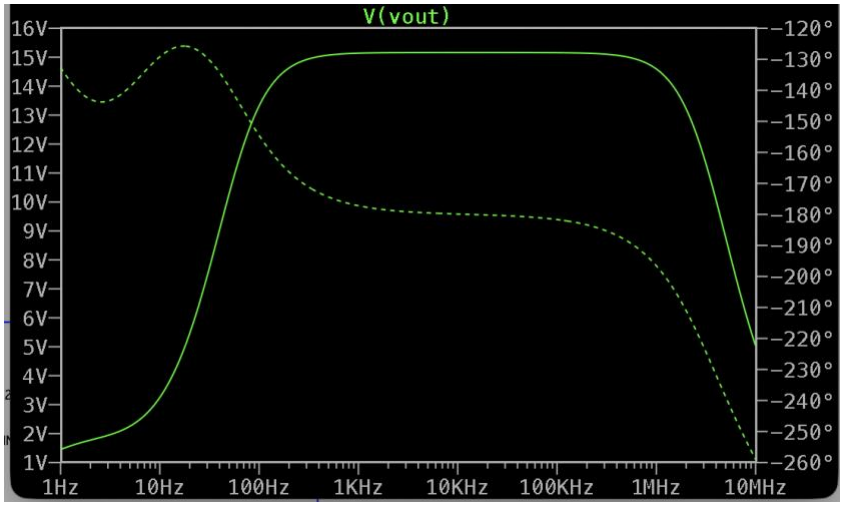


Figure 5. LTspice AC analysis result shown in linear voltage scale

The lower cutoff frequency is determined by the large capacitors in the circuit. These are the input coupling capacitor, the output coupling capacitor, and the source bypass capacitor. Each one creates a low-frequency pole, and the largest of these pole frequencies usually dominates the lower cutoff frequency.

For the input coupling capacitor, the resistance seen by the capacitor is mainly the input resistance of the bias network. R1 and R2 are in parallel, so the input resistance is approximately 66.7 kΩ. This gives a very small cutoff frequency, about 0.024 Hz.

For the output coupling capacitor, the resistance seen by the capacitor is approximately RD + RL, which is 2 kΩ. This gives a cutoff frequency of about 0.8 Hz.

For the source bypass capacitor, the resistance seen by the capacitor is much smaller because it is related to 1/gm in parallel with RS. This gives the largest low-frequency pole, around 57.5 Hz. Since this is the largest of the three low-frequency values, it dominates the lower cutoff frequency. Therefore:

fL ≈ 57.5 Hz

This means the amplifier starts to lose gain below this frequency mainly because the source bypass capacitor no longer acts like a perfect short circuit.

The upper cutoff frequency is mainly determined by the internal capacitances of the BS170 transistor. At high frequencies, these parasitic capacitances start to affect the gain. The gate-drain capacitance is especially important because it is affected by the Miller effect. In the handwritten calculation, the high-frequency poles are estimated by finding the equivalent input and output capacitances. The input high-frequency pole is found to be around 22.2 MHz, while the output pole is around 7.8 MHz. Since the smaller pole dominates, the upper cutoff frequency is approximately:

fH ≈ 7.8 MHz

The hand-calculated upper cutoff frequency was estimated using a simplified capacitance model and was found to be approximately 7.8 MHz. However, the LTspice AC analysis shows that the gain reaches the upper 3-dB level earlier, around 3.5 MHz. This difference is expected because the BS170 VDMOS model includes non-ideal and nonlinear parasitic capacitances, internal resistances, and other effects that are not included in the simplified hand calculation. Therefore, the simulated f_H is lower than the theoretical estimate.

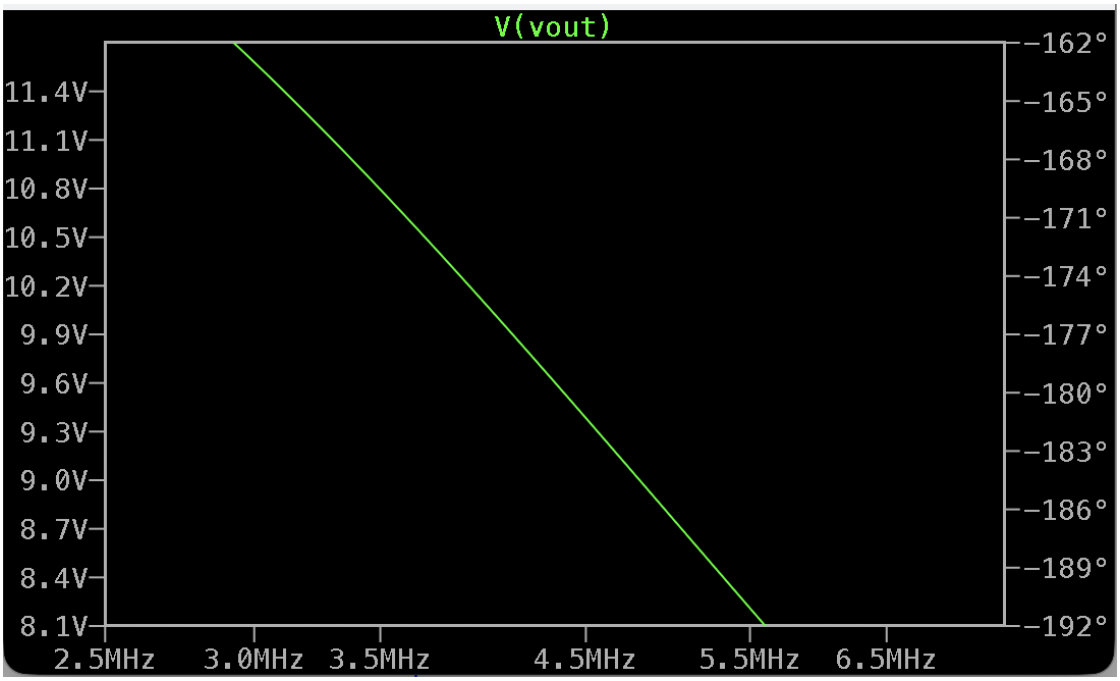


Figure 6. Zoomed-in high-frequency response of the amplifier showing the upper 3-dB cutoff region and the decrease in gain at high frequencies.

In the LTspice simulation, the circuit is built using the BS170 model provided in the assignment. The .op simulation is used to find the DC operating point. The simulation gives approximately ID = 4.12 mA, VGS = 2.34 V, and VDS = 4.89 V. These values are close to the hand calculations, although they are not exactly the same. The difference is expected because the LTspice BS170 model is more realistic than the simplified equations used in hand analysis. The model includes internal resistances, nonlinear capacitances, and other non-ideal transistor effects.

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Circuit: *
Start Time: Thu May 14 21:52:29 2026
solver = Normal
Maximum thread count: 10
tnom = 27
temp = 27
method = modified trap
CompressWinPoints = 1024
Direct Newton iteration for .op point succeeded.
Semiconductor Device Operating Points:
                        --- VDMOSFET Transistors ---
Name:          m1
Model:        bs170
Id:          4.12e-03
Vgs:         2.34e+00
Vds:         4.89e+00
Vth:         2.08e+00
Gm:          3.19e-02
Gds:         0.00e+00
Cgs:         2.80e-11
Cgd:         5.48e-12
Cbody:       1.44e-11


Total elapsed time: 0.034 seconds.
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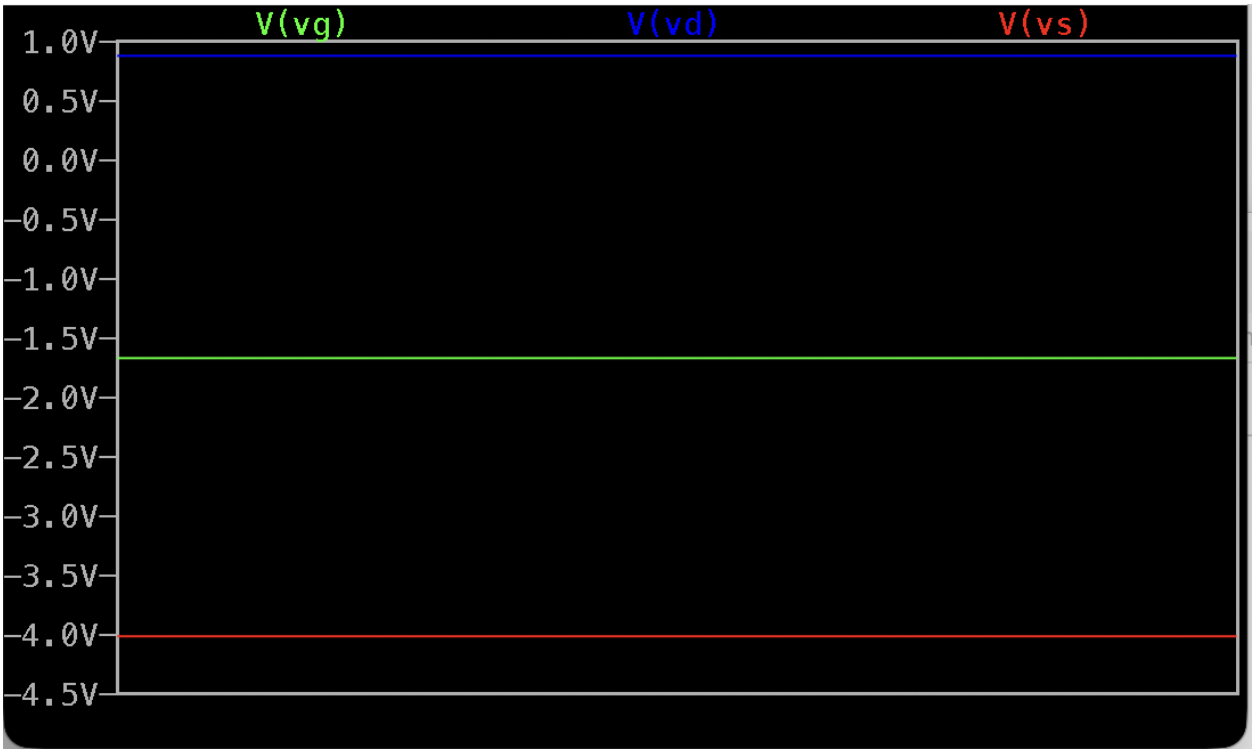


Figure 7. DC operating points.

Overall, the common-source amplifier is biased properly in the saturation region. The calculated Q-point is approximately IDQ = 4.43 mA and VDSQ = 4.5 V. The DC terminal voltages are VG = −1.67 V, VS = −3.94 V, and VD = 0.568 V. The DC load line goes from 10 V at zero current to 8.06 mA at zero VDS, and the Q-point lies on this line.

In the small-signal analysis, the amplifier gives an inverted output with a midband gain of about −15.6 V/V. The lower cutoff frequency is mainly controlled by the source bypass capacitor and is approximately 57.5 Hz. The upper cutoff frequency is mainly controlled by the transistor parasitic capacitances and is approximately 7.8 MHz. The LTspice results are close to the hand calculations, but small differences are normal because the simulation uses a more detailed and realistic BS170 model.

3. Data/Results

Part 1: Determination of Conduction Parameter and Threshold Voltage

Building Block: Diode-Connected BS170 NMOS Measurement Circuit

In this part of the experiment, the BS170 small-signal enhancement-mode n-channel MOSFET was tested in a diode-connected configuration. The purpose of this part was to experimentally determine the threshold voltage, VT, and the conduction parameter, Kn, of the transistor. To do this, the gate and drain terminals of the MOSFET were connected together, as shown in the lab manual. With this connection, the gate-to-source voltage becomes equal to the drain-to-source voltage, so VGS = VDS. This setup makes it possible to obtain two operating points of the MOSFET and use them to calculate VT and Kn.

The source terminal of the transistor was connected to the common reference point. A 1 kΩ resistor was connected between the positive terminal of the DC power supply and the drain terminal of the MOSFET.

The DC supply voltage was then adjusted to obtain two different drain current values. For each case, both the drain current and the corresponding gate-to-source voltage were measured.

Components:

BS170 NMOS transistor
1 kΩ resistor
DC power supply
Digital multimeter
Breadboard and connection wires

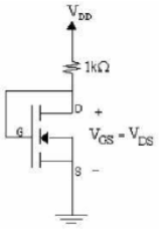


Figure 8. Schematic of the diode-connected BS170 NMOS measurement circuit.

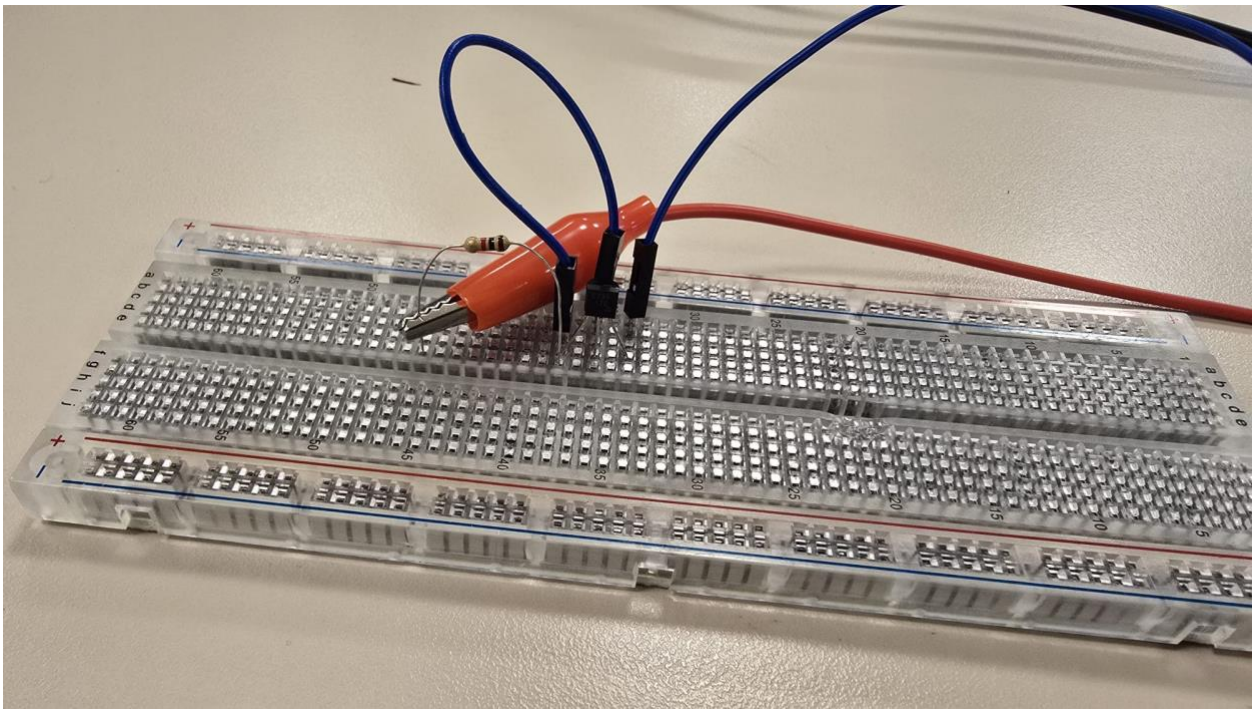
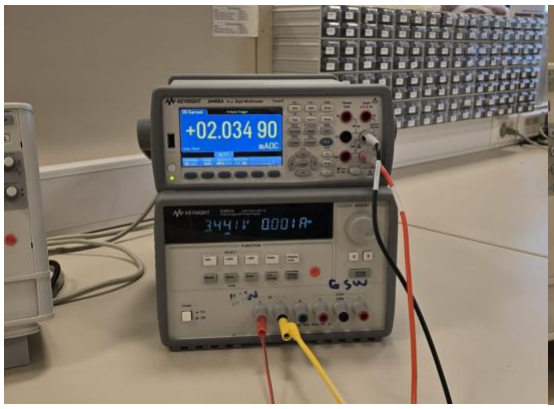


Figure 9. Breadboard setup used for determining VT and Kn of the BS170 NMOS transistor.

Measurement

For the first measurement, the DC supply voltage was adjusted until the voltage drop across the 1 kΩ resistor was approximately 2 V. Since the resistor value was 1 kΩ, this corresponds to a drain current close to 2 mA. The measured drain current was 2.035 mA. At this same operating point, the gate-to-source voltage was measured as 1.435 V.

ID1 = 2.035 mA
VGS1 = 1.435 V



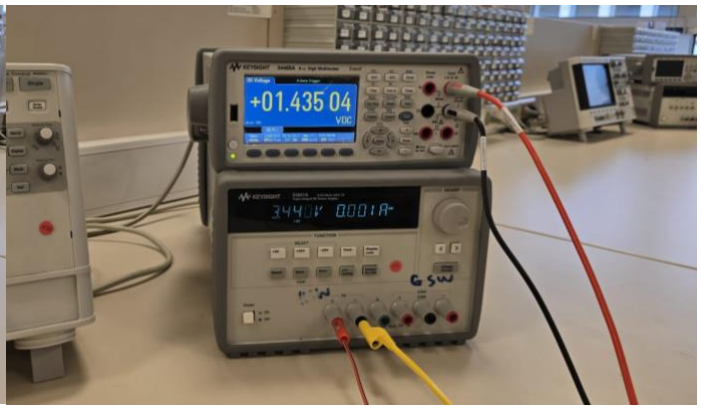


Figure 10. Multimeter measurement of the first drain current value, ID1 = 2.035 mA. And Figure 11. Multimeter measurement of the first gate-to-source voltage, VGS1 = 1.435 V.

For the second measurement, the DC supply voltage was increased until the voltage drop across the 1 kΩ resistor was approximately 4 V. This gave a drain current close to 4 mA. The measured drain current was 4.000 mA, and the corresponding gate-to-source voltage was measured as 1.604 V.

ID2 = 4.000 mA
VGS2 = 1.604 V

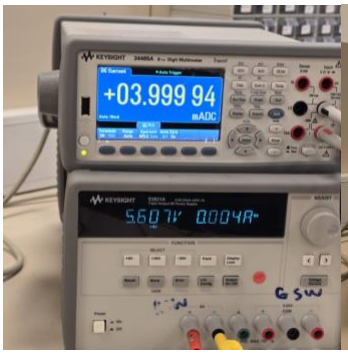




Figure 12. Multimeter measurement of the second drain current value, ID2 = 4.000 mA. And Figure 13. Multimeter measurement of the second gate-to-source voltage, VGS2 = 1.604 V.

Using the two measured operating points, the threshold voltage was calculated with the equation given in the lab manual:

$$V_T = (V_{GS2} - V_{GS1}\sqrt{(I_{D2} / I_{D1})}) / (1 - \sqrt{(I_{D2} / I_{D1})})$$

Substituting the measured values:

$$V_T = (1.604 - 1.435\sqrt{(4.000 / 2.035)}) / (1 - \sqrt{(4.000 / 2.035)})$$

$$V_T \approx 1.02 \text{ V}$$

After finding the threshold voltage, the conduction parameter was calculated using the first measurement point:

Kn = ID1 / (VGS1 − VT)^2
Kn = 2.035 mA / (1.435 − 1.02)^2
Kn ≈ 11.5 mA/V²

To check whether this extracted Kn value was consistent with the second measurement, ID2 was calculated again using VGS2, VT, and Kn:

ID2 = Kn(VGS2 − |VT|)^2
ID2 = 11.5(1.604 − 1.02)^2
ID2 ≈ 3.922 mA

The calculated value of ID2 was 3.922 mA, while the measured value was 4.000 mA. These two values are very close to each other. The small difference is most likely caused by resistor tolerance, multimeter reading tolerance, slight changes in the supply adjustment, and the non-ideal behavior of the actual MOSFET.

Percent difference = |4.000 − 3.922| / 4.000 × 100

Percent difference ≈ 1.95%

The percent difference was only about 1.95%, so the calculated values of VT and Kn are consistent with the experimental measurements. Therefore, the threshold voltage of the BS170 MOSFET was found to be approximately 1.02 V, and the conduction parameter was found to be approximately 11.5 mA/V².

Summary of Part 1 Measurements

Quantity	Measured / Calculated Value
ID1	2.035 mA
VGS1	1.435 V
ID2	4.000 mA
VGS2	1.604 V
VT	1.02 V
Kn	11.5 mA/V²
ID2 from calculation	3.922 mA
Percent difference for ID2	1.95%

Table1.

Overall, this part of the experiment successfully characterized the BS170 NMOS transistor using two measured operating points. The calculated ID2 value was very close to the measured ID2 value, which shows that the extracted VT and Kn values were reasonable for this transistor. Since actual MOSFET parameters can vary from device to device, the experimentally obtained values were not expected to be exactly the same as the ideal values used in the theoretical pre-lab analysis.

Part 2: Common-Source Amplifier Implementation

Building Block: Common-Source BS170 NMOS Amplifier

In this part of the experiment, the common-source amplifier circuit was built using the BS170 NMOS transistor. The purpose of this part was to implement the amplifier analyzed in the pre-lab, measure its DC operating point, observe the input and output waveforms, calculate the voltage gain, and examine the frequency response of the amplifier.

The circuit was constructed using +5 V and −5 V supply voltages. The gate terminal was biased by a voltage divider formed by R1 and R2. The drain terminal was connected to +5 V through RD1, and the source terminal was connected to −5 V through RS1. The input signal was applied through the coupling capacitor CC, and the output signal was taken from the drain through the coupling capacitor CL across the load resistor RL. The source bypass capacitor CS was connected in parallel with RS1 to increase the AC gain by making the source terminal approximately AC ground at midband frequencies.

This configuration is called a common-source amplifier because the source terminal is common to the input and output signals for AC operation. The output is taken from the drain, so the output signal is expected to be amplified and inverted with respect to the input signal. The lab manual requires building the circuit, measuring the DC voltages at the gate, source, and drain terminals, applying a 1 kHz sinusoidal input signal, measuring the voltage gain, and then taking frequency response data for the Bode plot.

Components:

BS170 NMOS transistor
R1 = 200 kΩ resistor
R2 = 100 kΩ resistor
RD1 = 1 kΩ resistor
RS1 = 240 Ω resistor
RL = 1 kΩ resistor
CC = 100 µF capacitor
CL = 100 µF capacitor
CS = 100 µF capacitor
DC power supply
Function generator
Oscilloscope
Digital multimeter
Breadboard and connection wires

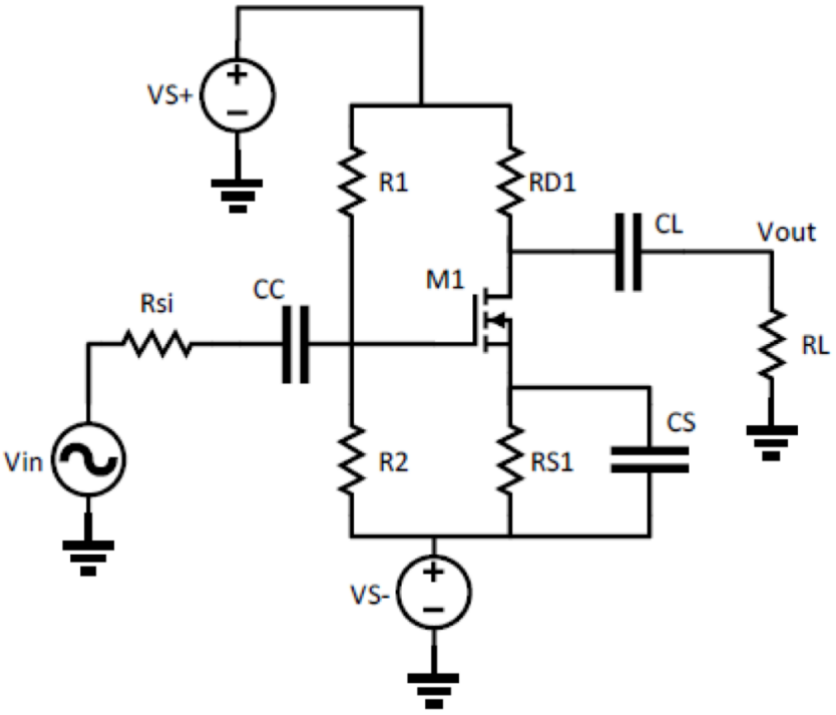


Figure 14. Schematic of the common-source BS170 NMOS amplifier circuit.

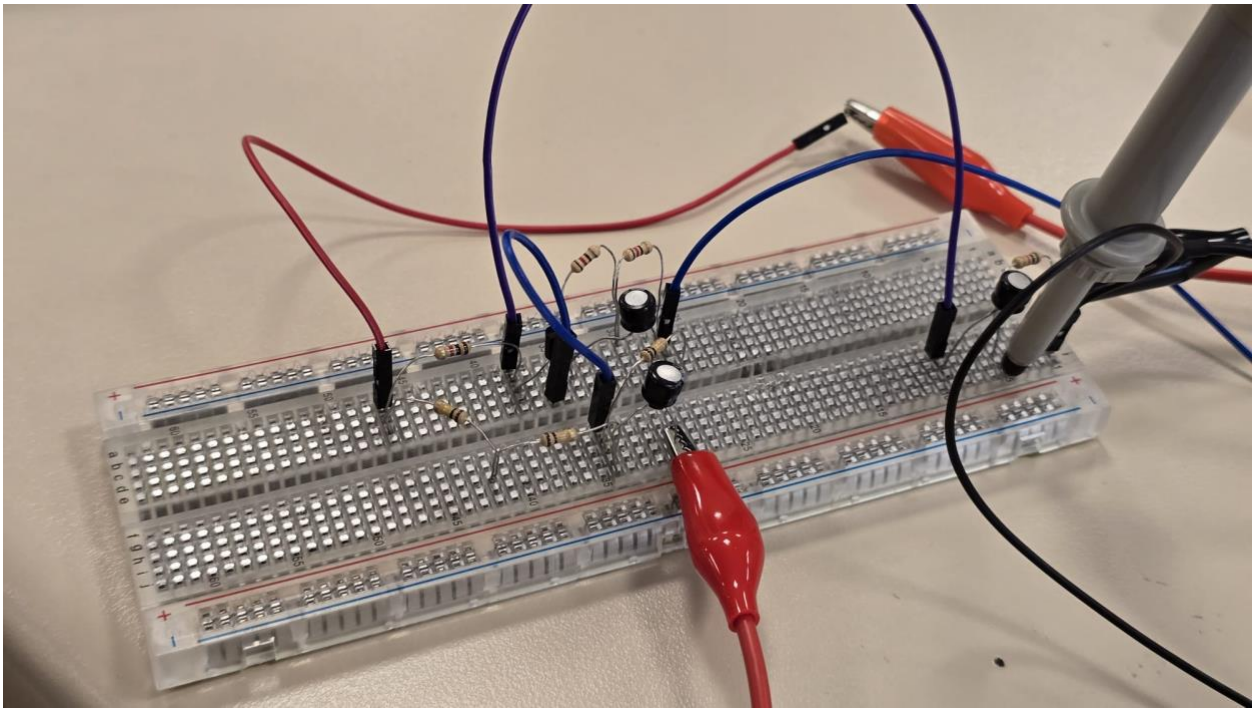


Figure 15. Breadboard implementation of the common-source amplifier circuit.

DC Voltage Measurements

After the amplifier circuit was built, the DC voltages at the gate, source, and drain terminals were measured with respect to the common reference point. These measurements were taken before applying the AC input signal. The purpose of these measurements was to check whether the transistor was biased close to the expected Q-point.

The theoretical/calculated values from the pre-lab were compared with the measured values. The measured gate voltage was very close to the calculated value because the gate voltage is mainly determined by the resistor divider. The source and drain voltages showed more noticeable differences, which is expected because the actual BS170 transistor parameters can vary from the simplified values used in the theoretical calculation.

Node	Measured Voltage	Calculated Voltage	Error Percentage
Gate	−1.67 V	−1.67 V	0%
Source	−3.50 V	−3.94 V	11.6%
Drain	1.171 V	0.568 V	45.5%

Table 2.

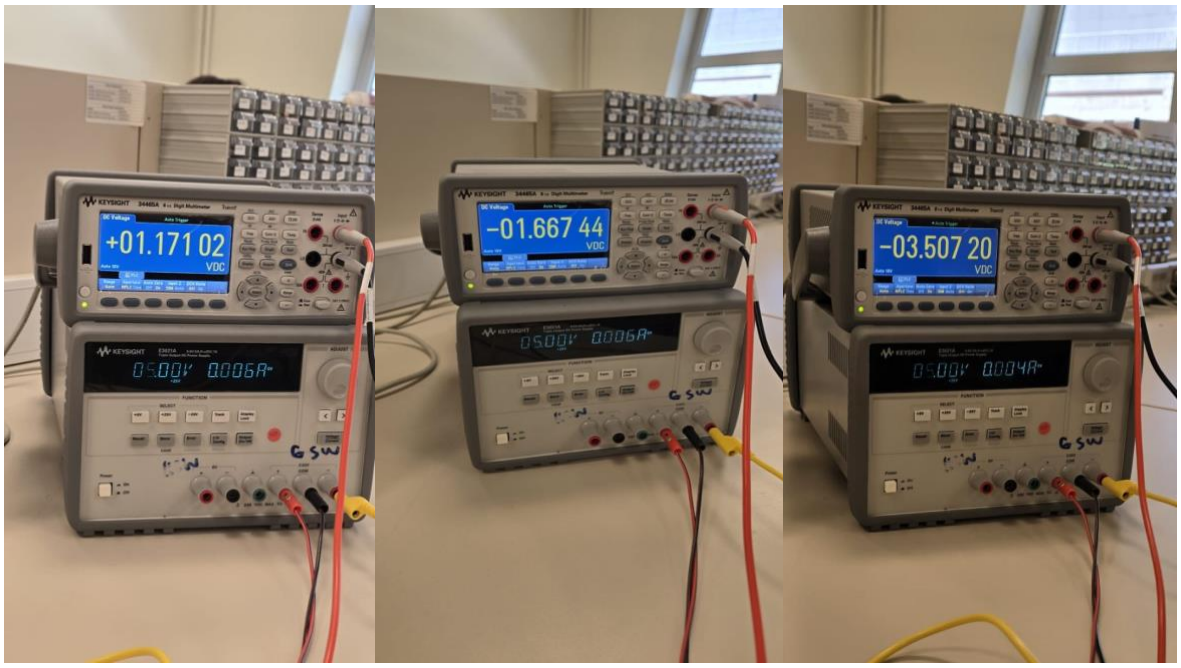


Figure 16. Measured drain voltage of the common-source amplifier, VD = 1.171 V. Figure 17. Measured gate voltage of the common-source amplifier, VG = −1.67 V. Figure 18. Measured source voltage of the common-source amplifier, VS = −3.50 V.

The gate voltage matched the calculated value almost exactly, which shows that the voltage divider bias was working properly. The source voltage was slightly less negative than the theoretical value, and the drain voltage was higher than the theoretical value. This indicates that the actual drain current was lower than the ideal theoretical current. Since MOSFET parameters such as threshold voltage and conduction parameter are not identical for every device, these differences are reasonable in a real circuit.

AC Gain Measurement at 1 kHz

After the DC bias was checked, a sinusoidal input signal was applied to the amplifier using the function generator. The input frequency was set to 1 kHz, which is in the midband region of the amplifier. The input amplitude was adjusted to 50 mVpp to avoid clipping or distortion at the output.

The oscilloscope was used to observe the output waveform. At 1 kHz, the output waveform remained sinusoidal, meaning the amplifier was operating in its linear region. The measured output peak-to-peak voltage was approximately 1.05 Vpp.

Vin,pp = 50 mVpp
Vout,pp = 1.05 Vppi

The voltage gain was calculated by dividing the output peak-to-peak voltage by the input peak-to-peak voltage:

Av = Vout,pp / Vin,pp. Av = 1.05 V / 0.050 V. Av ≈ 21 V/V

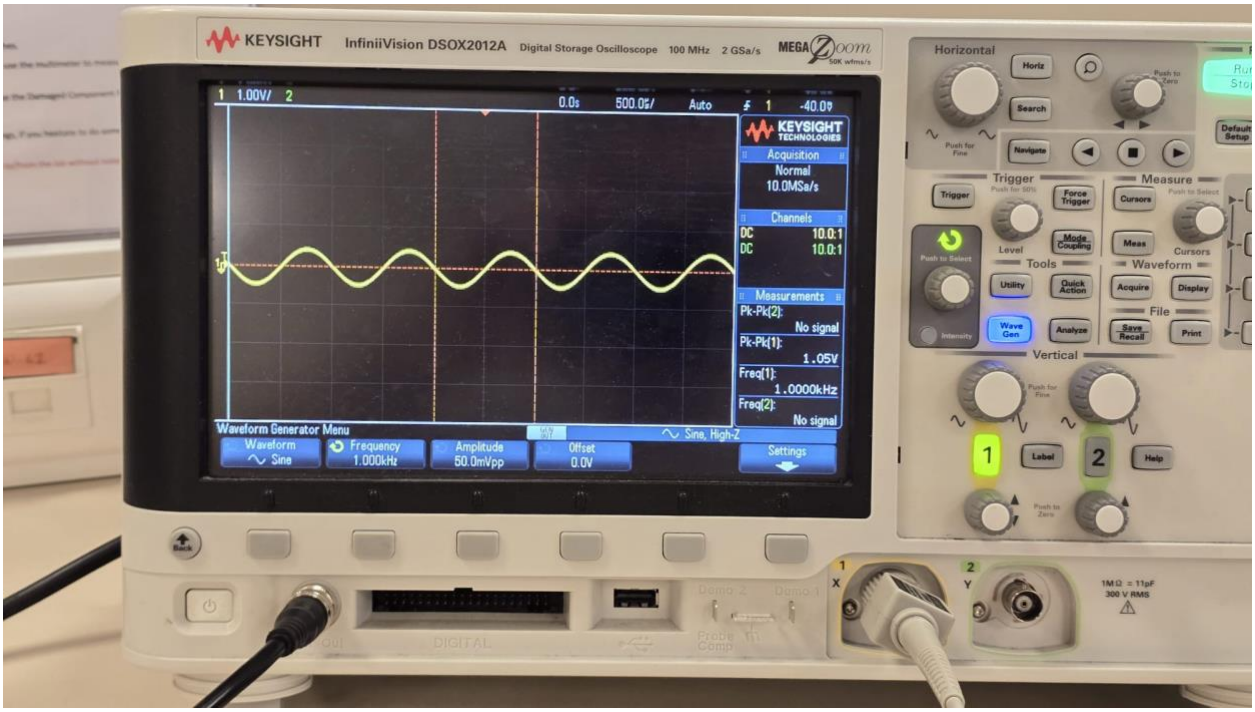


Figure 19. Oscilloscope measurement of the output waveform at 1 kHz with 50 mVpp input signal.

The experimentally obtained voltage gain was approximately 21 V/V. The pre-lab theoretical gain was approximately 15.6 V/V. The measured gain was higher than the theoretical value. This difference can be caused by the actual BS170 transistor having different small-signal parameters than the simplified

theoretical model. Also, component tolerances, capacitor behavior, breadboard parasitic effects, and measurement conditions can change the practical gain of the amplifier.

The percent difference between theoretical and experimental gain is:
Percent difference = |21 − 15.6| / 15.6 × 100
Percent difference ≈ 34.6%

Although there is a difference between the theoretical and measured gain, the amplifier still behaved as expected for a common-source amplifier. It produced a larger output signal from a small input signal, and the output waveform remained sinusoidal without visible clipping.

Frequency Response Measurement

To observe the frequency response of the amplifier, the input signal amplitude was kept at 50 mVpp while the frequency was changed. At each frequency, the output peak-to-peak voltage was measured with the oscilloscope. These measurements will be used to calculate the gain at each frequency and to draw the Bode plot of the amplifier.

Some example frequency response measurements were observed during the experiment. Around 58.95 Hz, the output voltage decreased to approximately 720 mVpp. At a high-frequency point around 1.403 MHz, the output voltage was approximately 700 mVpp. Since the midband output was about 1.05 Vpp, these lower output values show that the gain decreases at both low and high frequencies.

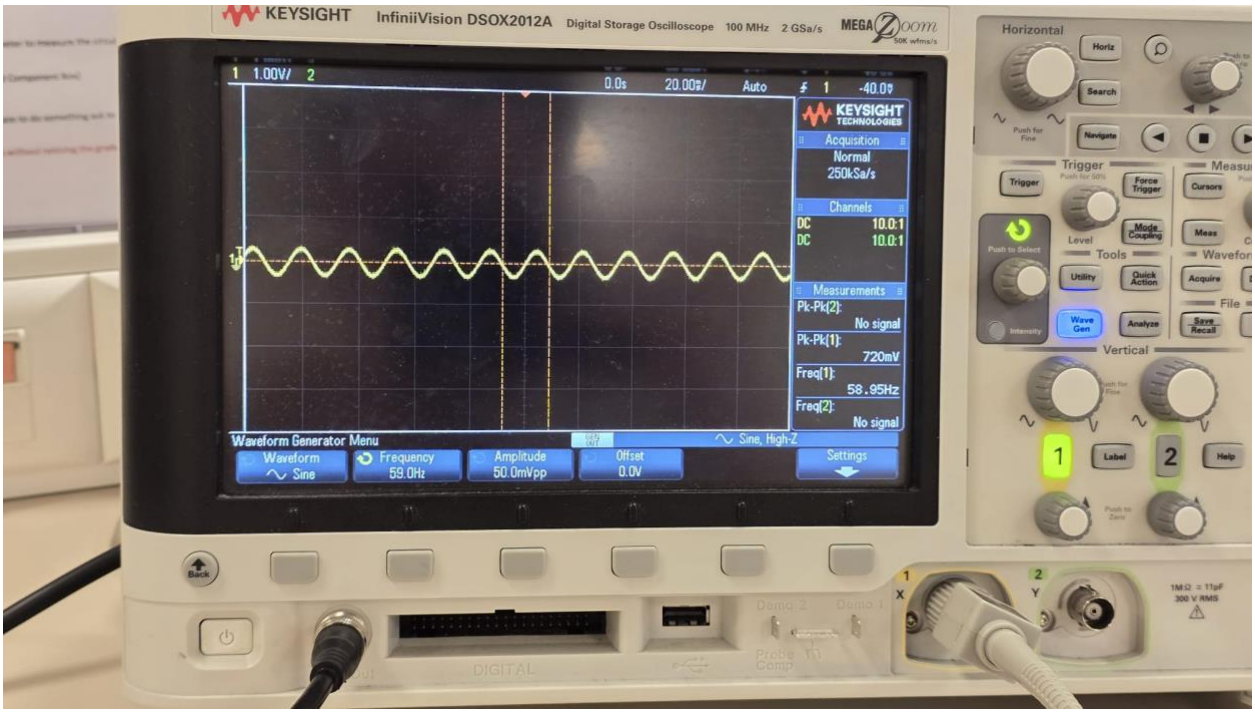


Figure 20. Low-frequency output measurement at approximately 58.95 Hz.

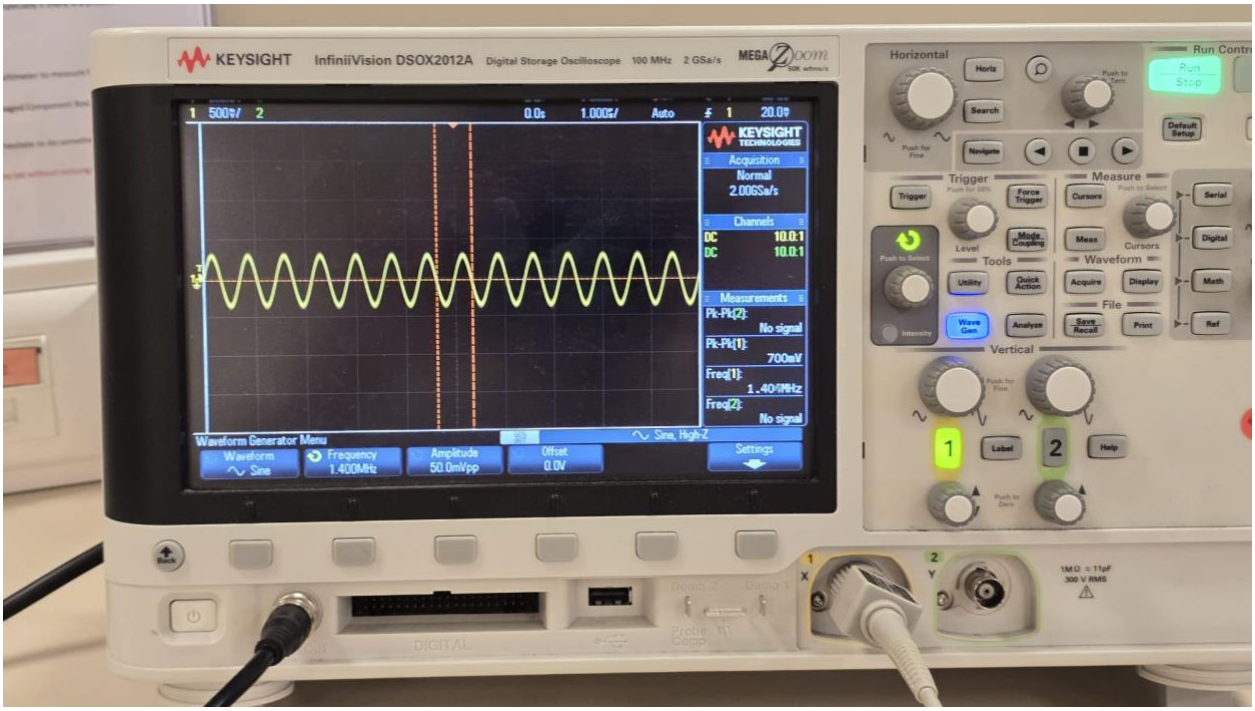
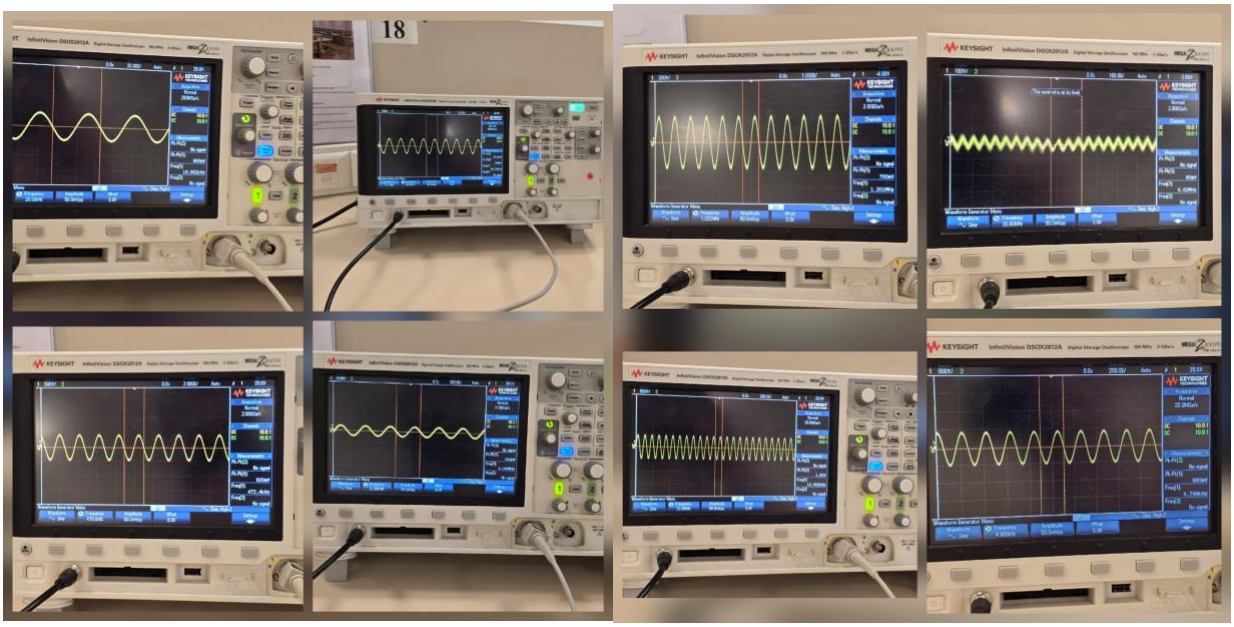


Figure 21. High-frequency output measurement at approximately 1.403 MHz.

The decrease at low frequencies is mainly caused by the coupling capacitors and the source bypass capacitor. At low frequencies, these capacitors no longer behave like ideal short circuits, so the gain decreases. The decrease at high frequencies is mainly caused by the internal capacitances of the MOSFET and parasitic capacitances in the breadboard circuit. These capacitances become more significant at high frequencies and reduce the amplifier gain.

The full frequency response table and Bode plot will be completed using the measured frequency and output voltage values. From this Bode plot, the lower cutoff frequency fL and upper cutoff frequency fH can be estimated by finding the frequencies where the gain drops by 3 dB from the midband gain.



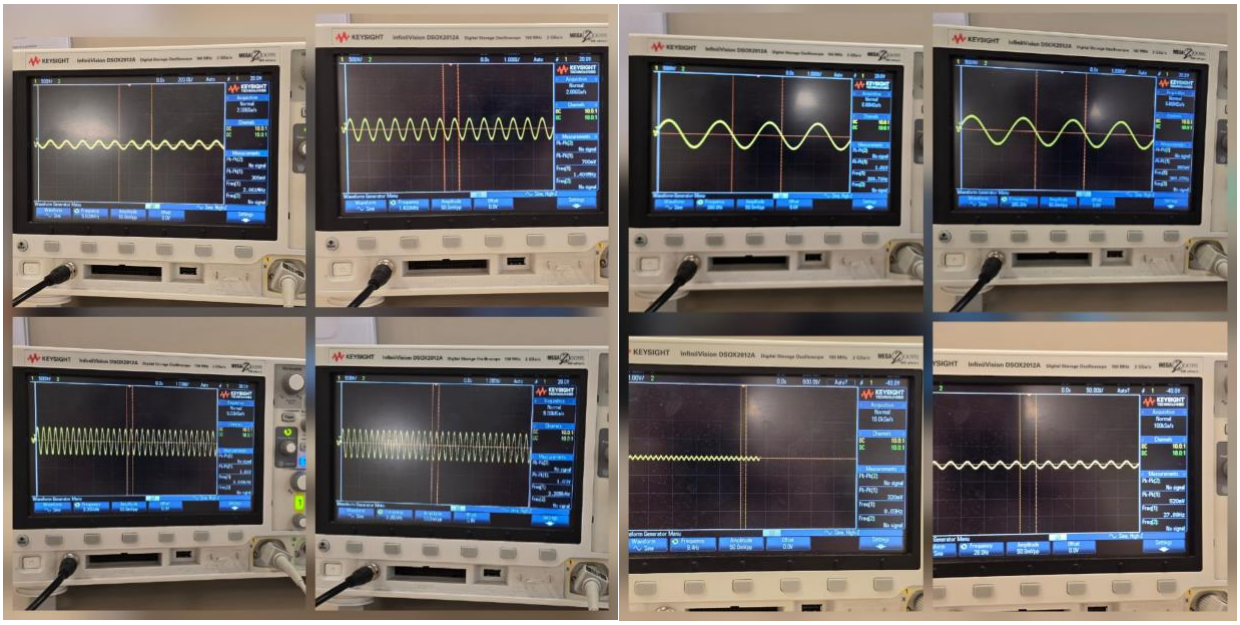


Figure 22. Additional oscilloscope measurements used to complete the frequency response data. These screenshots show the output waveform behavior at different low, midband, and high frequency values.

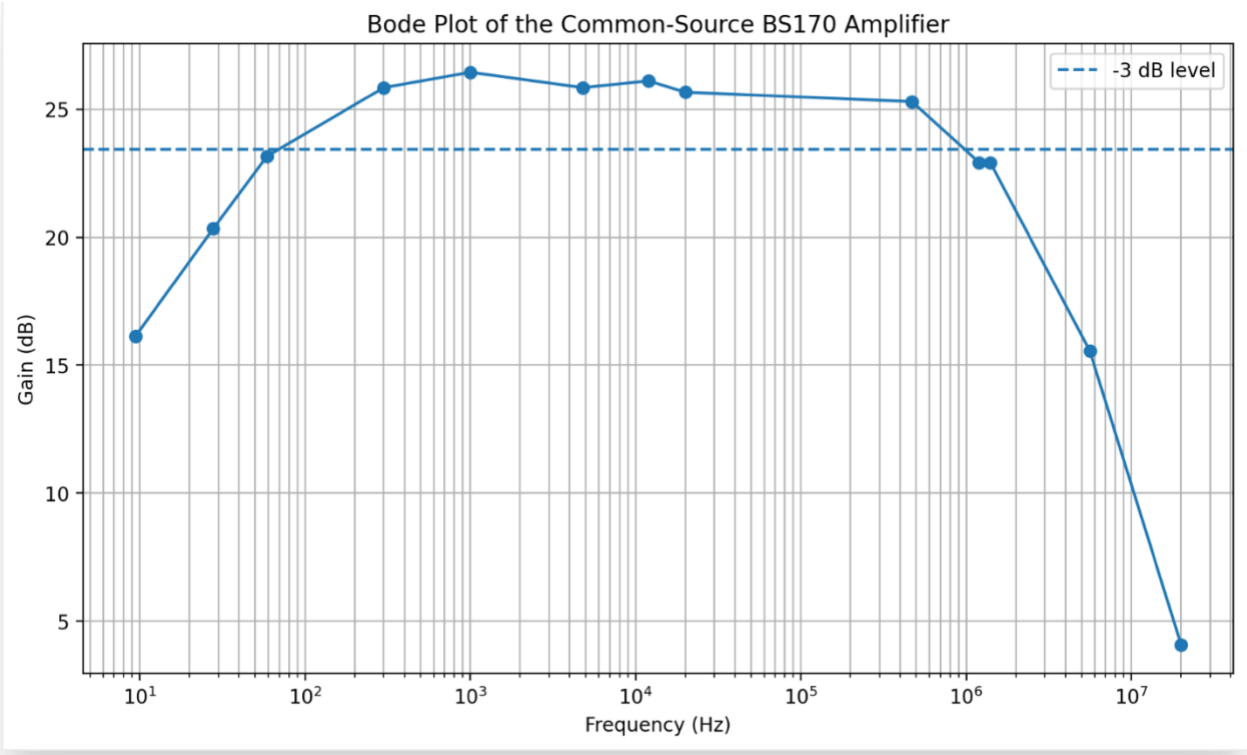


Figure 23. Bode plot of the common-source BS170 NMOS amplifier. The gain was calculated from the measured output voltage values by using Av = Vout,pp / Vin,pp, where Vin,pp was kept at 50 mVpp. The dashed line represents the −3 dB level.

The frequency response of the common-source amplifier was measured by keeping the input signal amplitude constant at 50 mVpp and changing the input frequency over a wide range. For each frequency, the output peak-to-peak voltage was recorded from the oscilloscope and used to calculate the voltage gain. The Bode plot shows that the amplifier has a relatively constant gain in the midband region, where the gain is approximately 21 V/V, or about 26 dB. At low frequencies, the gain decreases because the coupling capacitors and the source bypass capacitor no longer behave like ideal short circuits. At high frequencies, the gain also decreases due to the internal capacitances of the MOSFET and parasitic capacitances in the breadboard circuit. From the plot, the lower cutoff frequency is around 60 Hz, and the upper cutoff frequency is approximately in the MHz range. This behavior is consistent with the expected frequency response of a capacitor-coupled common-source MOSFET amplifier.

Summary of Part 2 Measurements

Quantity	Measured Value
VG	−1.67 V
VS	−3.50 V
VD	1.171 V
Input frequency for gain measurement	1 kHz
Vin,pp	50 mVpp
Vout,pp	1.05 Vpp
Experimental gain	21 V/V
Theoretical gain	15.6 V/V
Gain percent difference	34.6%

Table 3.

Overall, the common-source amplifier was successfully implemented on the breadboard. The measured DC voltages showed that the MOSFET was biased and operating as an amplifier. The AC measurement at 1 kHz showed that a small input signal of 50 mVpp was amplified to about 1.05 Vpp, giving an experimental gain of approximately 21 V/V. The gain was higher than the theoretical value, but the amplifier still showed the expected common-source behavior. The frequency response measurements also showed that the gain decreases at low and high frequencies, which is consistent with the expected behavior of a capacitor-coupled MOSFET amplifier.

4. Discussion/Analysis

In this experiment, the BS170 NMOS transistor was first characterized experimentally, and then it was used in a common-source amplifier circuit. The main results obtained from the experiment were generally consistent with the expected behavior of a MOSFET amplifier. In Part 1, the threshold voltage and conduction parameter were extracted from two measured operating points. The measured values were ID1 = 2.035 mA, VGS1 = 1.435 V, ID2 = 4.000 mA, and VGS2 = 1.604 V. Using these values, the threshold voltage was calculated as VT ≈ 1.02 V and the conduction parameter was calculated as Kn ≈ 11.5 mA/V². When ID2 was recalculated using these extracted values, it was found as 3.922 mA, which is very close to the measured value of 4.000 mA. This small difference shows that the parameter extraction was successful.

For the common-source amplifier part, the measured DC terminal voltages were compared with the calculated values from the pre-lab. The gate voltage was measured as −1.67 V, which matched the calculated value almost exactly. This was expected because the gate voltage is mainly determined by the R1–R2 voltage divider. However, the measured source voltage and drain voltage were different from the calculated values. The source voltage was measured as −3.50 V instead of −3.94 V, and the drain voltage was measured as 1.171 V instead of 0.568 V. These differences indicate that the actual drain current of the transistor was lower than the theoretical value predicted in the pre-lab.

Quantity	**Theoretical Value**	**Measured Value**	**Percent Error**
VG	−1.67 V	−1.67 V	0%
VS	−3.94 V	−3.50 V	11.6%
VD	0.568 V	1.171 V	45.5%
Voltage gain	15.6 V/V	21 V/V	34.6%
Lower cutoff frequency	≈ 57.5 Hz	≈ 60 Hz	close
Upper cutoff frequency	≈ 7.8 MHz theoretical / lower in practice	≈ 1.1–1.3 MHz	different

Table 4.

The largest error was observed in the drain voltage. This is reasonable because the drain voltage directly depends on the drain current flowing through RD. Even a moderate difference in the actual drain current can create a noticeable change in the drain voltage. The main reason for this difference is that the real BS170 transistor does not behave exactly like the simplified theoretical model. MOSFET parameters such as threshold voltage, conduction parameter, internal resistance, and capacitances can vary from one transistor to another. In addition, resistor tolerances, breadboard contact resistance, and measurement limitations can also affect the result.

The voltage gain measurement also showed a difference between theory and experiment. The theoretical midband gain was about 15.6 V/V, while the measured gain at 1 kHz was approximately 21 V/V. Although the measured gain was higher than the theoretical value, the circuit still behaved as a common-source amplifier. The input was 50 mVpp, and the output was approximately 1.05 Vpp without significant clipping or distortion. This shows that the amplifier was operating in the linear region. The difference in gain can be explained by the actual small-signal transconductance of the MOSFET, component tolerances, and the fact that the theoretical calculation uses a simplified model.

The frequency response results were also consistent with the expected behavior of a capacitor-coupled MOSFET amplifier. In the midband region, the gain stayed relatively constant around 21 V/V, or about 26 dB. At low frequencies, the gain decreased because the input coupling capacitor, output coupling capacitor, and source bypass capacitor no longer behaved like ideal short circuits. This caused signal attenuation and reduced the amplifier gain. The measured lower cutoff frequency was around 60 Hz, which is very close to the theoretical lower cutoff frequency of about 57.5 Hz.

At high frequencies, the gain also decreased. This decrease was mainly caused by the internal capacitances of the BS170 MOSFET and parasitic capacitances from the breadboard and wires. The measured upper cutoff frequency was around the MHz range, approximately 1.1–1.3 MHz from the Bode plot. This was lower than the simplified theoretical estimate. This difference is expected because the high-frequency theoretical calculation does not include all non-ideal effects of the real transistor and experimental setup.

Overall, the results of the experiment were reasonable and showed the expected operation of a common-source MOSFET amplifier. The circuit successfully amplified a small sinusoidal input signal, and the measured frequency response showed both low-frequency and high-frequency gain reduction. The discrepancies between theoretical and measured values were mainly caused by real transistor parameter variation, non-ideal capacitances, resistor tolerances, breadboard parasitics, and measurement limitations. These differences show why practical circuit measurements are important in addition to theoretical calculations and simulations.

5. Post-Lab

In this experiment, the behavior of a BS170 NMOS transistor and its use in a common-source amplifier circuit were examined experimentally. The lab consisted of two main parts. In the first part, the threshold voltage and conduction parameter of the MOSFET were extracted from measurement data. In the second part, a common-source amplifier was implemented, and its DC operating point, voltage gain, and frequency response were measured. The overall results showed that the circuit behaved as expected, but some differences were observed between theoretical predictions and experimental measurements. These differences are important because they show the limitations of ideal circuit analysis when working with real electronic components.

In the first part, the MOSFET was connected in a diode-connected configuration by shorting the gate and drain terminals. In this configuration, VGS and VDS are equal. This is useful because the transistor can be

forced to operate at different current levels, and the corresponding gate-to-source voltages can be measured. Using two different operating points, the threshold voltage VT and conduction parameter Kn were calculated. The measured values were ID1 = 2.035 mA, VGS1 = 1.435 V, ID2 = 4.000 mA, and VGS2 = 1.604 V. From these values, VT was found to be approximately 1.02 V, and Kn was found to be approximately 11.5 mA/V^2.

The threshold voltage is the minimum gate-to-source voltage needed for the MOSFET to begin conducting significantly. However, in a real MOSFET, this value is not fixed for every device. It can vary due to manufacturing differences, temperature, and measurement conditions. The conduction parameter Kn represents how strongly the transistor conducts after it turns on. A higher Kn means that for the same overdrive voltage, VGS − VT, the transistor allows more drain current. Since these parameters directly affect the drain current, they also affect the DC bias point and voltage gain of the amplifier.

After calculating VT and Kn, the second drain current was recalculated using the equation $ID = Kn(VGS - VT)^2$. The calculated current was 3.922 mA, while the measured value was 4.000 mA. These values were very close to each other, with only a small percent difference. This showed that the extracted values of VT and Kn were reasonable for the actual BS170 transistor used in the lab. It also confirmed that the measurement method was successful.

In the second part, the BS170 transistor was used in a common-source amplifier configuration. In this amplifier, the input signal is applied to the gate terminal, and the output is taken from the drain terminal. The source terminal is connected through a source resistor, and for AC signals, the source bypass capacitor makes the source node approximately AC ground in the midband frequency range. Because of this, the amplifier provides voltage gain. A common-source amplifier also inverts the signal, meaning that the output waveform is approximately 180 degrees out of phase with the input waveform.

The DC operating point is very important in this circuit because the MOSFET must be biased properly before it can amplify an AC signal. If the transistor is not biased correctly, the output waveform may be clipped or distorted. In this lab, the gate voltage was set by the voltage divider formed by R1 and R2. The measured gate voltage was −1.67 V, which was almost the same as the theoretical value. This was expected because the gate current of a MOSFET is extremely small, so the voltage divider determines the gate voltage quite accurately.

However, the measured source and drain voltages were different from the theoretical values. The source voltage was measured as −3.50 V, while the theoretical value was about −3.94 V. The drain voltage was measured as 1.171 V, while the theoretical value was about 0.568 V. These differences show that the actual drain current was different from the value predicted by the simplified theoretical model. Since the drain voltage is found from the voltage drop across the drain resistor, any change in drain current directly changes the drain voltage. This explains why the drain voltage had a larger percent error compared to the other node voltages.

One of the main reasons for this difference is that the theoretical analysis uses ideal or simplified MOSFET parameters. In reality, the BS170 transistor does not behave exactly according to the simple square-law model. The actual threshold voltage and conduction parameter may be different from the values assumed in pre-lab calculations. Also, the transistor model may include internal resistances and nonlinear effects that are not included in hand calculations. Resistor tolerance, breadboard contact

resistance, and small measurement errors can also contribute to the difference between calculated and measured values.

The voltage gain measurement was performed at 1 kHz with an input amplitude of 50 mVpp. The output was approximately 1.05 Vpp, giving an experimental gain of about 21 V/V. This result showed that the amplifier successfully amplified the input signal. The output waveform was sinusoidal and did not show significant clipping, which means the circuit was operating in a proper linear region. The theoretical gain was approximately 15.6 V/V, so the measured gain was higher than the predicted value. This difference can again be explained by the actual MOSFET parameters and the limitations of the simplified small-signal model.

The voltage gain of a common-source amplifier mainly depends on the transconductance gm and the effective drain resistance. In the midband region, the coupling and bypass capacitors behave almost like short circuits. Therefore, the source resistor is bypassed for AC signals, and the gain becomes larger. If the actual gm of the transistor is different from the calculated gm, the measured gain will also be different. Since gm depends on the drain current and transistor parameters, even small differences in the DC operating point can affect the final gain.

The frequency response measurement showed that the amplifier had a limited bandwidth. In the midband region, the gain stayed relatively constant. At low frequencies, the gain decreased. This happens because the input coupling capacitor, output coupling capacitor, and source bypass capacitor have larger reactance at low frequencies. When the capacitive reactance is high, the capacitors cannot pass the AC signal effectively. As a result, part of the signal is attenuated, and the voltage gain decreases. The measured lower cutoff frequency was around 60 Hz, which was close to the theoretical lower cutoff frequency of approximately 57.5 Hz. This agreement shows that the low-frequency behavior was mainly determined by the external capacitors, as expected.

At high frequencies, the gain also decreased. This high-frequency drop is caused by the internal capacitances of the MOSFET, such as gate-source and gate-drain capacitances. The gate-drain capacitance is especially important because of the Miller effect. In an inverting amplifier, the effective input capacitance becomes larger due to the voltage gain, which reduces the high-frequency response. In addition to transistor capacitances, breadboard parasitic capacitance, long wires, oscilloscope probe capacitance, and non-ideal connections can also reduce the gain at high frequencies. Because of these effects, the measured upper cutoff frequency was lower than the simplified theoretical estimate.

The Bode plot clearly demonstrated the expected band-pass-like behavior of the amplifier. The gain increased from low frequencies to the midband region, stayed almost constant over a range of frequencies, and then decreased at high frequencies. This is typical for a capacitor-coupled common-source amplifier. The −3 dB points indicate the lower and upper cutoff frequencies of the amplifier. These cutoff frequencies are important because they define the usable bandwidth of the circuit. In practical amplifier design, the bandwidth must be selected according to the frequency range of the signal that needs to be amplified.

The discrepancies observed in this experiment have important implications for practical electronic circuit design. In real circuits, theoretical calculations are necessary for initial design, but they are not enough by themselves. Actual component values may not be exactly equal to their nominal values. MOSFET parameters can vary significantly between different devices even if they have the same part number.

Also, high-frequency behavior is strongly affected by parasitic capacitances and physical layout. For this reason, practical circuits should be tested experimentally and, when necessary, adjusted after measurement.

Overall, this experiment showed that the BS170 MOSFET can be used successfully as a common-source amplifier. The circuit amplified a small sinusoidal input signal, produced a stable output waveform, and showed the expected frequency-dependent gain behavior. The measured results were not exactly identical to the theoretical predictions, but the differences were reasonable and explainable. The experiment demonstrated the importance of biasing, small-signal analysis, transistor parameter extraction, and frequency response measurement in MOSFET amplifier design. It also showed that combining theoretical analysis, simulation, and laboratory measurements gives a more complete understanding of real electronic circuit performance.

6. Conclusion

In this experiment, the BS170 NMOS transistor was examined and used to build a common-source amplifier circuit. The first part of the lab focused on finding the threshold voltage and conduction parameter of the MOSFET experimentally. By using two different operating points, the threshold voltage was calculated as approximately 1.02 V, and the conduction parameter was calculated as approximately 11.5 mA/V^2. The recalculated drain current was very close to the measured current, which showed that the extracted transistor parameters were reasonable.

In the second part of the lab, the common-source amplifier was implemented on a breadboard. The DC terminal voltages at the gate, source, and drain were measured and compared with the theoretical values. The gate voltage matched the expected value very closely, while the source and drain voltages showed some differences. These differences were mainly caused by the non-ideal behavior of the real BS170 MOSFET, variation in transistor parameters, resistor tolerances, and practical measurement conditions.

The amplifier successfully increased the amplitude of a small sinusoidal input signal. With a 50 mVpp input signal at 1 kHz, the output voltage was measured as approximately 1.05 Vpp, giving a voltage gain of about 21 V/V. Although this gain was different from the theoretical value, the circuit still behaved as expected for a common-source amplifier. The output waveform remained sinusoidal and did not show significant clipping, which indicated that the MOSFET was biased in a suitable operating region.

The frequency response measurement showed that the amplifier had a limited bandwidth. The gain was almost constant in the midband region, but it decreased at both low and high frequencies. The low-frequency drop was mainly caused by the coupling and bypass capacitors, while the high-frequency drop was caused by MOSFET internal capacitances and parasitic effects from the breadboard and wires. This confirmed the expected frequency-dependent behavior of a capacitor-coupled MOSFET amplifier.

Overall, the experiment was successful because the measured results supported the main theoretical expectations. The common-source amplifier provided voltage amplification, showed phase-inverting amplifier behavior, and had clear lower and upper cutoff regions in its Bode plot. The differences between theoretical and experimental values were reasonable and showed the importance of considering real component behavior in practical circuit design. This lab helped demonstrate that

theoretical calculations, simulations, and experimental measurements should be used together to understand and design reliable electronic amplifier circuits.